

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (Civil) (Credit System) Degree Examinations**

April 2023

21CV303 – BUILDING MATERIALS AND ENGINEERING GEOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain origin, texture, structures, engineering properties and uses of any one common Igneous rock.	5	L*2	1	1,2
b)	Explain rock cycle with a neat sketch.	5	L2	1	1,2
c)	Explain any 5 tests to be conducted to check the quality of bricks as per BIS.	10	L2	2	1,2
2. a)	Examine the internal structure and composition of the earth with a neat sketch.	10	L2	1	1,2
b)	Explain the properties and uses of the following: i) Aluminum ii) Brass iii) Cast iron iv) Mild steel	10	L2	2	1
3. a)	Explain with suitable examples, how will you identify a mineral with its physical properties depending upon light.	10	L2	1	1,2
b)	Explain any 7 characteristics to check the quality of a good timber. What are the two basic types of tiles?	10	L2	2	1
Unit – II					
4. a)	How to decide the location of doors in a building? Define the following terms with respect to stairs: i) Flight ii) Landing iii) Headroom iv) Going v) Scotia	10	L2	3	1
b)	Explain the types of Folds based on attitude of axial plane and degree of compression with neat sketches.	10	L2	4	1,2
5. a)	List the different types of windows. Explain any 2 types of windows with a neat sketch.	10	L2	3	1
b)	Explain different parts of a normal fault with a neat sketch.	10	L2	4	1,2
6. a)	What are the different types of flooring? Explain any 2 types of flooring in detail.	10	L2	3	1
b)	Explain the significance of joints in civil engineering.	5	L2	4	1,2,4,5
c)	Demonstrate clinometer compass and its uses with a neat sketch.	5	L2	4	1,2
Unit – III					
7. a)	What is pile foundation? Explain in detail (sketch) the different classification of piles with respect to load transmission.	10	L2	5	1,2
b)	Explain the types and causes of landslides.	10	L2	5	1,2
8. a)	Explain the dropping weight method to determine the bearing capacity of soil with a neat sketch. What are the essential requirements of a good foundation?	10	L2	5	1
b)	Formulate the engineering considerations to be taken for building seismic resistant structures.	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
